



Lesson 7.9 — Trigonometric Identities

Big idea: Identities turn one trig expression into another. The Pythagorean identities are the backbone — almost everything else follows from them.

Quick review

Pythagorean identities: $\sin^2\theta + \cos^2\theta = 1$ (the main one). Dividing through gives $1 + \tan^2\theta = \sec^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$.

Reciprocal: $\csc\theta = 1/\sin\theta$, $\sec\theta = 1/\cos\theta$, $\cot\theta = 1/\tan\theta$.

Quotient: $\tan\theta = \sin\theta/\cos\theta$, $\cot\theta = \cos\theta/\sin\theta$.

Even/odd: $\sin(-\theta) = -\sin\theta$, $\cos(-\theta) = \cos\theta$, $\tan(-\theta) = -\tan\theta$.

Sum and double-angle (introductory): $\sin(2\theta) = 2 \sin\theta \cos\theta$; $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 1 - 2 \sin^2\theta = 2 \cos^2\theta - 1$.

Worked example

Worked example. Simplify $(\sin x)(1 + \cot^2 x)$.

$1 + \cot^2 x = \csc^2 x$. So expression = $\sin x \cdot \csc^2 x = \sin x / \sin^2 x = 1/\sin x = \csc x$.

Warm-ups

1. Simplify $\sin^2\theta + \cos^2\theta$.
2. Simplify $(1 - \cos^2\theta)$.
3. Simplify $\csc\theta \cdot \sin\theta$.

Practice

1. Simplify $(1 - \sin^2\theta)/\cos\theta$.
2. Simplify $(\sec\theta - \cos\theta)$.
3. Verify: $\tan\theta + \cot\theta = \sec\theta \csc\theta$.

Challenge

1. If $\sin\theta = 4/5$ and θ is in Q2, find $\cos\theta$, $\tan\theta$, $\sec\theta$, $\csc\theta$, $\cot\theta$ exactly.
2. Use a double-angle identity: if $\cos\theta = 1/3$, find $\cos(2\theta)$.



Answer key — Lesson 7.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

- 1.
- $\sin^2\theta$.
- 1.

Practice

- $\cos^2\theta/\cos\theta = \cos\theta$.
- $1/\cos\theta - \cos\theta = (1 - \cos^2\theta)/\cos\theta = \sin^2\theta/\cos\theta = \sin\theta \cdot \tan\theta$.
- $\tan\theta + \cot\theta = \sin\theta/\cos\theta + \cos\theta/\sin\theta = (\sin^2\theta + \cos^2\theta)/(\sin\theta \cos\theta) = 1/(\sin\theta \cos\theta) = \csc\theta \sec\theta$. ✓

Challenge

- $\cos\theta = -3/5$ (Q2). $\tan\theta = (4/5)/(-3/5) = -4/3$. $\sec\theta = -5/3$. $\csc\theta = 5/4$. $\cot\theta = -3/4$.
- $\cos(2\theta) = 2\cos^2\theta - 1 = 2(1/9) - 1 = -7/9$.